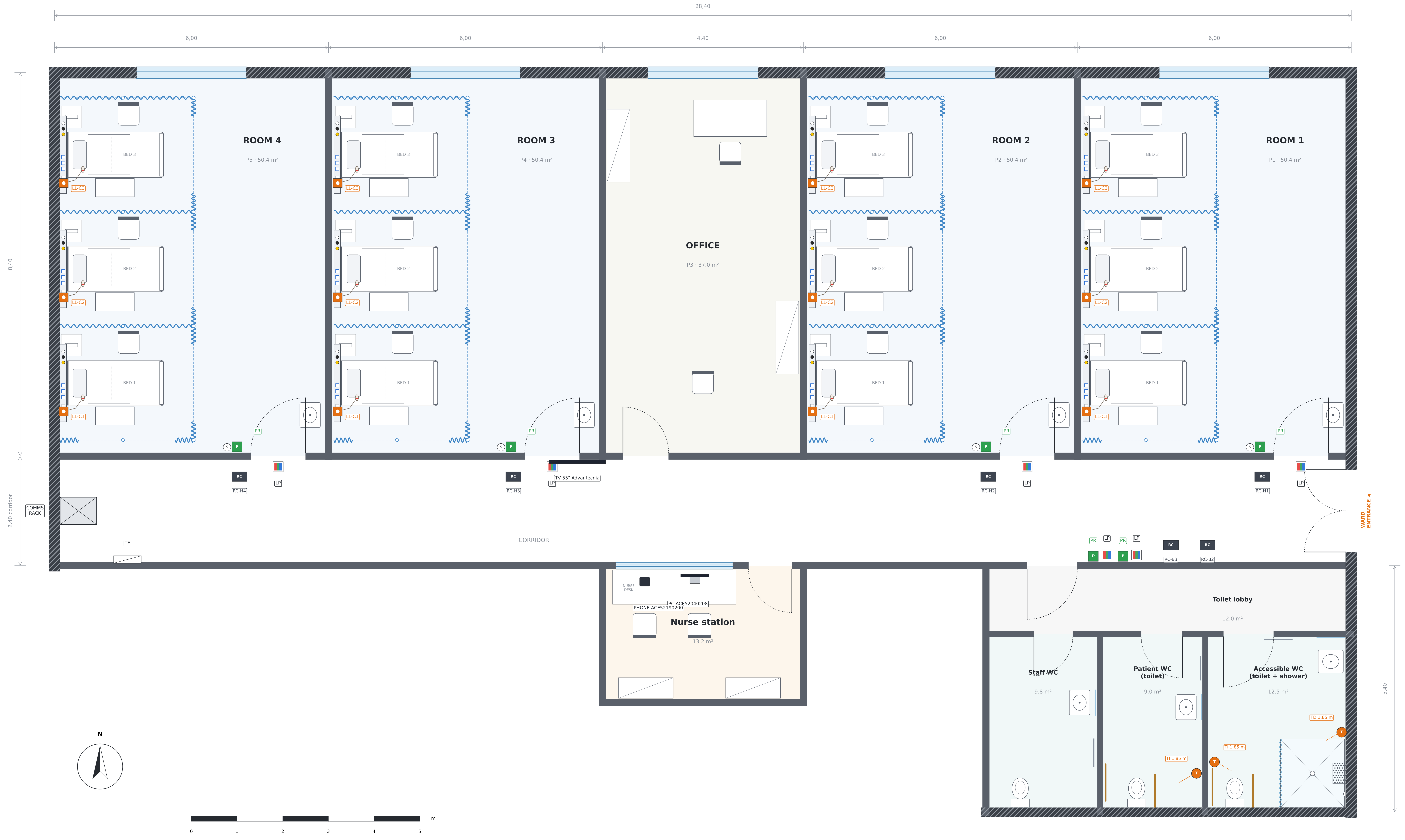


SYMBOLS

🟧	LL · Bed call ACE41220316 + pear push (1 per bed) Bed call panel + pear push · סגא + הטימל האירק ןצחל
🟥🟩🟦	LP · Corridor indicator light ACE41020309 (red call · green presence · blue code blue) Corridor indicator light, 3 colours · תיעבצ-תלת ןורדסמ תרונמ
RC	RC · Converter board / room controller ECV169 (corridor ceiling) Room controller ECV169 (corridor ceiling) · בקב רדח ECV169 (ןורדסמ תרקת)
S	S · Room light switches Room light switches · הרואת יגתמ
PC	Nurse desk: phone ACE52190200 / PC ACE52040208 and TV Touch all-in-one PC & TV running Advantecnia · תנכות םע TV ךסמו עגמ בשחמ Advantecnia
P	PR · Presence / cancel ACE80081001 Presence / cancel button · לוטיב / תוחכונ ןצחל
T	TI/TD · Toilet / shower pull-cord ACE40320303 (≥ 1.80 m) Pull-cord (mounted ≥ 1.80 m) · םיתורישב לבח תכישמ (≥1.80 מ')
○●●□□□	CT · Bed-head unit: O_2 · air · vacuum, 230 V sockets, RJ45 Bed-head unit: medical gases, power, data · הטימ תלוסנוק: םיזג, למשח, תרושקת
GAB	Rack: SmartHub, switch ACEUCM6302, server ACE51010103 Comms rack: SmartHub, switch, server · ןורא תרושקת: SmartHub, גתמ, תרש
TE	Ward electrical sub-panel Electrical sub-panel · למשח חול

NOTES

1. Rooms: 3 beds with the head against the side wall, one bed-head unit per bed and a U-shaped ceiling-track curtain per bed.

2. Inside, next to the door: light switches + presence/cancel (h = 1.05 / 1.35 m).

3. Square indicator light ACE41020309 above each door frame, facing the corridor (h = 2.42 m).

4. WC B2 and B3: pull-cords at ≥ 1.80 m. Lights and presence buttons of B2/B3 in the corridor, next to the toilet-block entrance.

5. Nurse station opposite P3 with its own door; desk against the corridor wall (window): the nurse faces the corridor and the TV between P3 and P4.

INPATIENT WARD · NURSE CALL SYSTEM

Sistema de llamada de enfermería · Pabellón de internación

System: ACE BG · Advantecnia Prepared by: Telab International

GENERAL FLOOR PLAN – EQUIPMENT

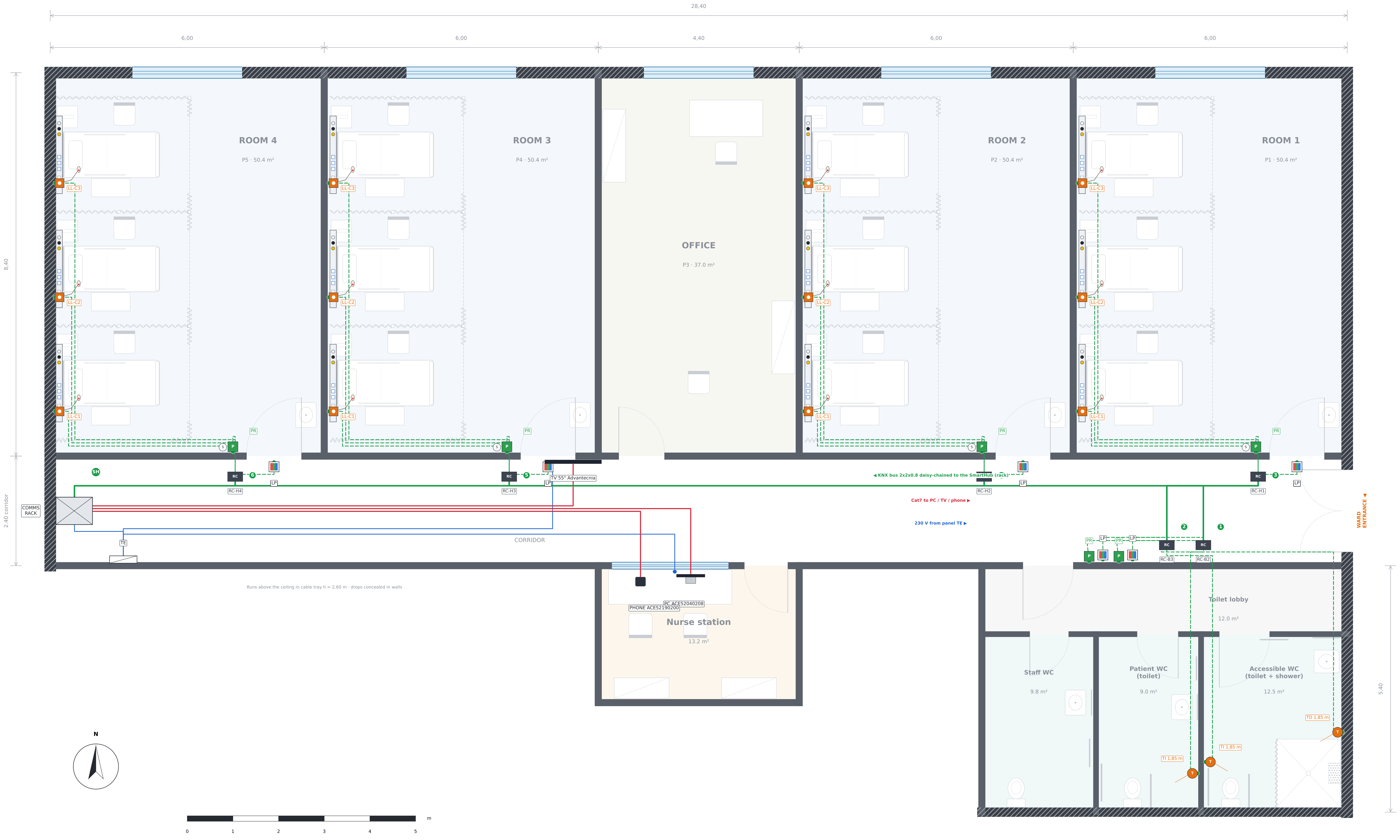
Planta general – equipamiento

םינקתהו דויצ – תיללכ תינכות

SCALE	DATE	REV	SHEET
1:50	12/10/2026	A	01

Dimensions in metres

A1 format · Rev. A – Preliminary design



CABLE COLOUR CODE

🟩	KNX bus 2x2x0.8 mm – 24-pin board → board (chain) → SmartHub KNX bus – board-to-board chain · BUS – שרשור כרטיסים
🟥	Cat6 / Cat7 only: SmartHub → switch → server · PC / phone · TV Cat6/Cat7 only: SmartHub, switch, server, PC/phone, TV · Cat6/Cat7 בלבד: SmartHub, מתג, שרת, מחשב, TV
- - -	Radial in each room – every device to its 24-pin board Radial in each room – device to its 24-pin board · רדיאלי בכל חדר – התקן לכרטיס
🟦	230 V power – rack, PC / phone and TV only (everything else is PoE) 230 V only to rack, PC/phone, TV – rest is PoE · 230V לארון, מחשב/טלפון ו-TV בלבד – השאר PoE

SYMBOLS

🟧	LL · Bed call ACE41220316 + pear push (1 per bed) Bed call panel + pear push · לחצן קריאה למיטה + אגס
🟥🟩🟦	LP · Corridor indicator light ACE41020309 (red call · green presence · blue code blue) Corridor indicator light, 3 colours · נורית מסדרון תלת-צבעית
RC	RC · Converter board / room controller ECV169 (corridor ceiling) Room controller ECV169 (corridor ceiling) · בקר חדר ECV169 (תקרת מסדרון)
S	S · Room light switches Room light switches · מתגי תאורה
PC	Nurse desk: phone ACE52190200 / PC ACE52040208 and TV Touch all-in-one PC & TV running Advantecnia · מחשב מגע מסך TV עם תוכנת Advantecnia
P	PR · Presence / cancel ACE80081001 Presence / cancel button · לחצן נוכחות / ביטול
T	TI/TD · Toilet / shower pull-cord ACE40320303 (≥ 1.80 m) Pull-cord (mounted ≥ 1.80 m) · משיכת חבל בשירותים (≥1.80 מ')
○●●☐☐☐	CT · Bed-head unit: O_2 · air · vacuum, 230 V sockets, RJ45 Bed-head unit: medical gases, power, data · קונסולת מיטה: גזים, חשמל, תקשורת
GAB	Rack: SmartHub, switch ACEUCM6302, server ACE51010103 Comms rack: SmartHub, switch, server · ארון תקשורת: SmartHub, מתג, שרת
TE	Ward electrical sub-panel Electrical sub-panel · לוח חשמל

NOTES

1. Each room is wired radially: every device (LL, PR, LP, TI) has its own cable to the room's 24-pin board (ECV169).

2. The 24-pin boards sit outside the room, above the corridor ceiling, and are chained board to board with KNX bus 2x2x0.8 mm LH EIBBUS.

3. The last RC (H4) ends the chain at the SmartHub (Raspberry) in the rack; then Cat6A to the switch and on to the server.

4. Cat6/Cat7 only for: SmartHub → switch → server, nurse PC or desk phone, and TV. No room has a Cat6/Cat7 connection.

5. The whole call system is PoE: 230 V only at the rack, PC / phone and TV.

INPATIENT WARD · NURSE CALL SYSTEM
Sistema de llamada de enfermería · Pabellón de internación
System: ACE BG · Advantecnia Prepared by: Telab International

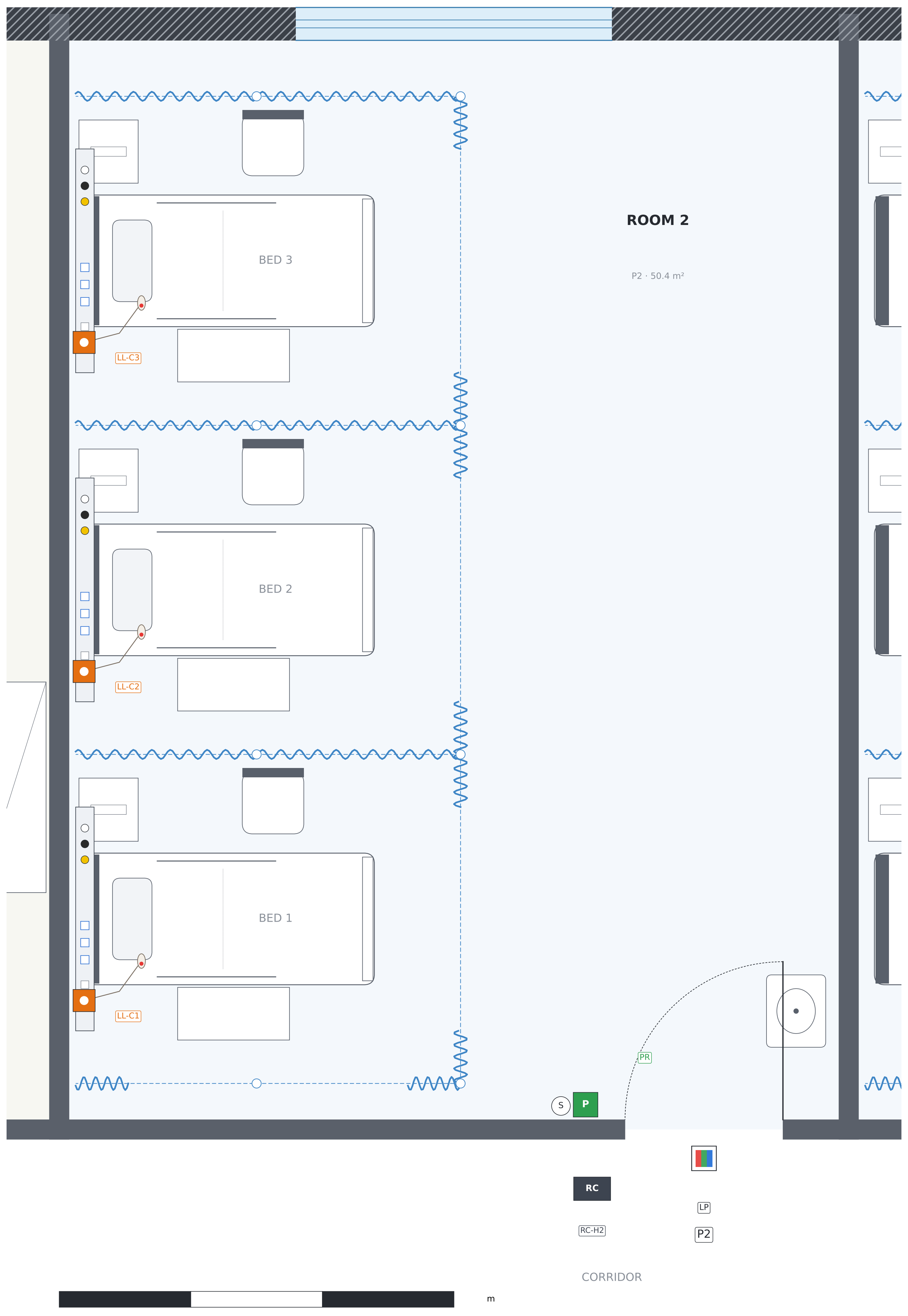
GENERAL FLOOR PLAN – WIRING

Planta general – cableado
תוכנית כללית – חיווט

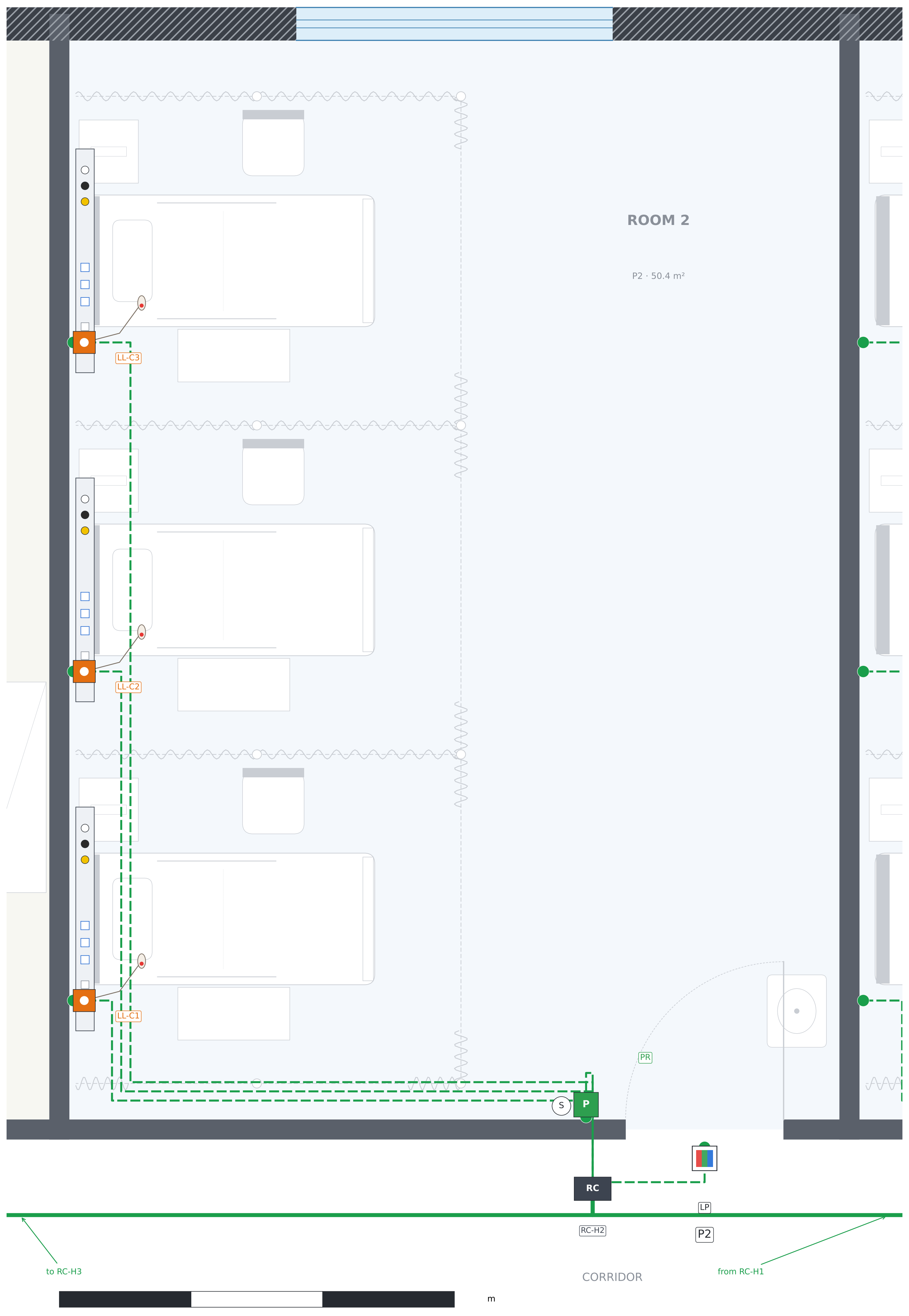
SCALE	DATE	REV	SHEET
1:50	12/10/2026	A	02

Dimensions in metres
A1 format · Rev. A – Preliminary design

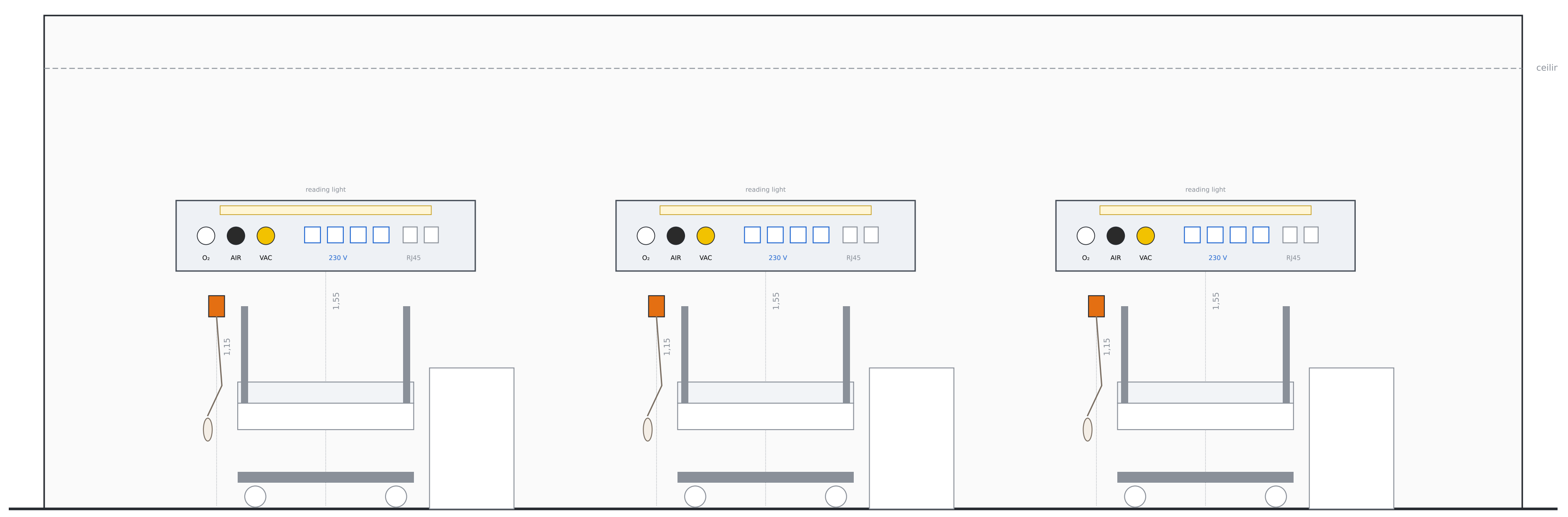
A · TYPICAL ROOM – EQUIPMENT · 1:30



B · TYPICAL ROOM – WIRING · 1:30

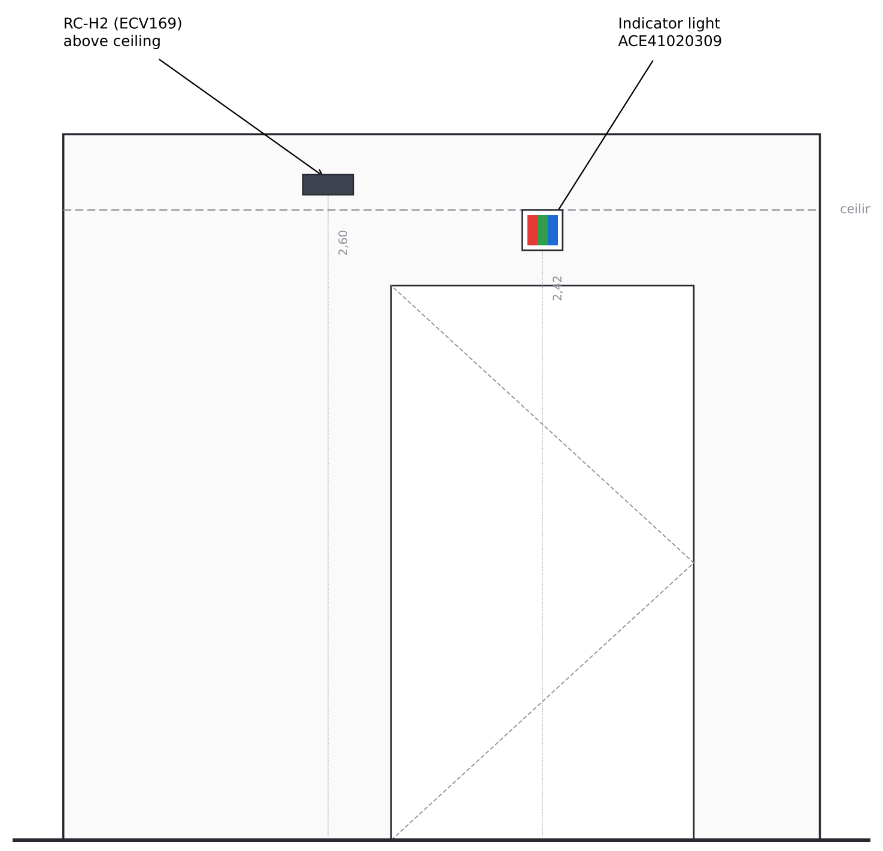


C · BED-HEAD WALL ELEVATION (from the room) · 1:30



View from the east

D · CORRIDOR ELEVATION – P2 · 1:30



View from the south

NOTES

1. Beds with the head against the side wall; bed-head unit per bed: O_2, air, vacuum, 4 × 230 V sockets, 2 RJ45, reading light.

2. Advantecnia bed call unit next to the bed-head (h = 1.15 m) with a detachable pear push reaching the bed.

3. Next to the door (inside): light switches h = 1.05 m and presence/cancel h = 1.35 m.

4. Indicator light ACE41020309 above the frame, facing the corridor, h = 2.42 m: red = call · green = presence · blue = code blue · off = no call.

5. 24-pin board ECV169 outside the room, above the corridor ceiling. Each LL, PR and LP arrives on its own cable (radial).

6. U-shaped curtain per bed: ceiling track on rods, white top mesh and light-blue flame-retardant fabric; front drawn back.

7. Window at the far end of each room, 2.40 × 1.30 m, sill at 0.90 m.

SYMBOLS

🟧	LL · Bed call ACE41220316 + pear push (1 per bed) Bed call panel + pear push · לחצן קריאה למיטה + אגס
🟥🟩🟦	LP · Corridor indicator light ACE41020309 (red call · green presence · blue code blue) Corridor indicator light, 3 colours · מנורת מסדרון תלת-צבעית
RC	RC · Converter board / room controller ECV169 (corridor ceiling) Room controller ECV169 (corridor ceiling) · בקר חדר ECV169 (תקרת מסדרון)
S	S · Room light switches Room light switches · מתגי תאורה
PC	Nurse desk: phone ACE52190200 / PC ACE52040208 and TV Touch all-in-one PC & TV running Advantecnia · מחשב מגע ומסך TV עם תוכנת Advantecnia
P	PR · Presence / cancel ACE80081001 Presence / cancel button · לחצן נוכחות / ביטול
T	TI/TD · Toilet / shower pull-cord ACE40320303 (≥ 1.80 m) Pull-cord (mounted ≥ 1.80 m) · משיכת חבל בשירותים (≥1.80 מ')
○●●□□□	CT · Bed-head unit: O_2 · air · vacuum, 230 V sockets, RJ45 Bed-head unit: medical gases, power, data · קונסולת מיטה: גזים, חשמל, תקשורת
GAB	Rack: SmartHub, switch ACEUCM6302, server ACE51010103 Comms rack: SmartHub, switch, server · ארון תקשורת: SmartHub, מתג, שרת
TE	Ward electrical sub-panel Electrical sub-panel · לוח חשמל

Heights in metres from finished floor. Advantecnia devices: LL 1.15 · PR 1.35 · light switches 1.05 · LP 2.42 · RC above ceiling.

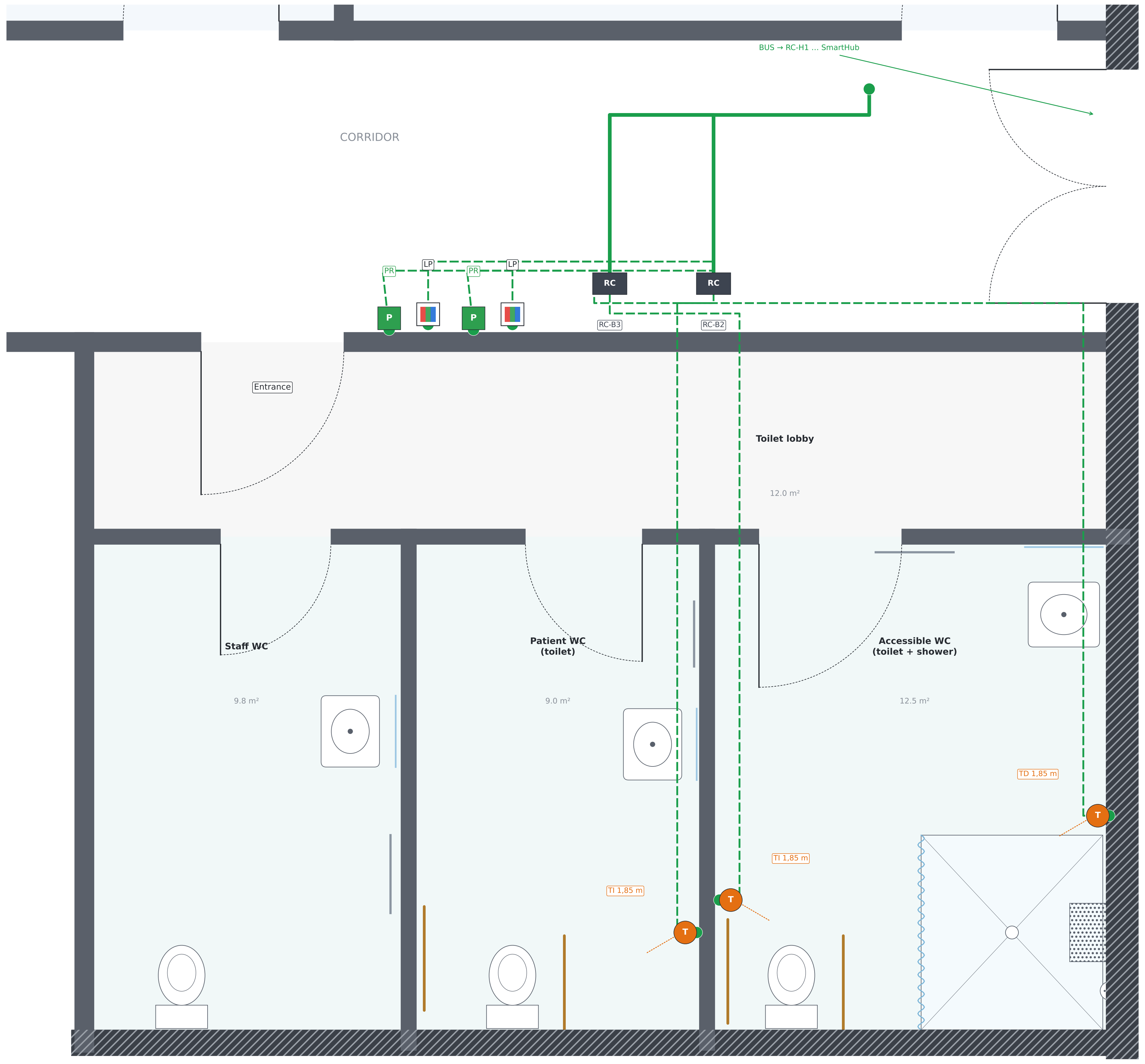
INPATIENT WARD · NURSE CALL SYSTEM
Sistema de llamada de enfermería · Pabellón de internación
System: ACE BG · Advantecnia Prepared by: Telab International

TYPICAL PATIENT ROOM (H2) – DETAIL
Habitación tipo – planta, cableado y alzados
חדר אשפוז טיפוסי – פרטים

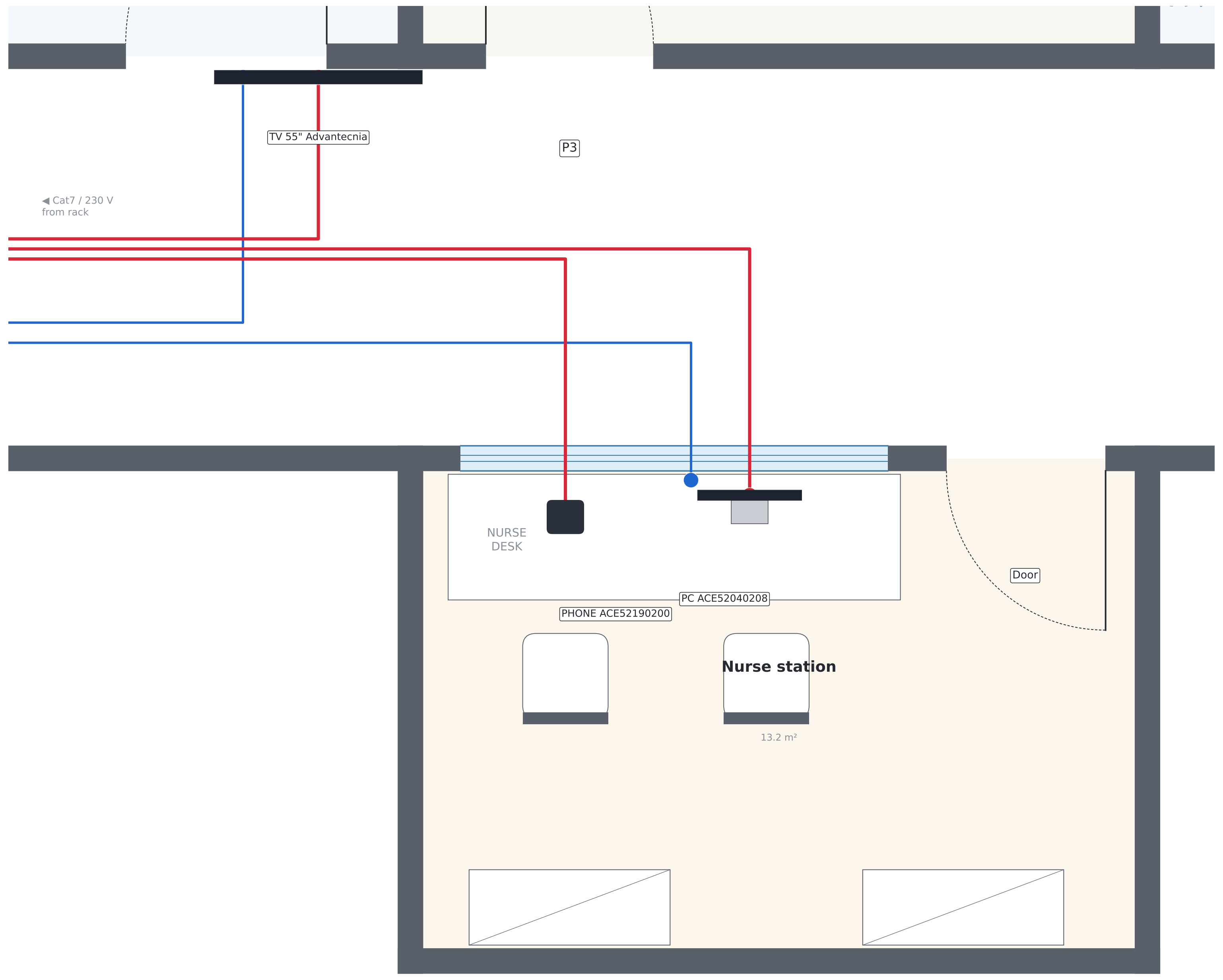
SCALE	DATE	REV	SHEET
1:30	12/10/2026	A	03

Dimensions in metres
A1 format · Rev. A – Preliminary design

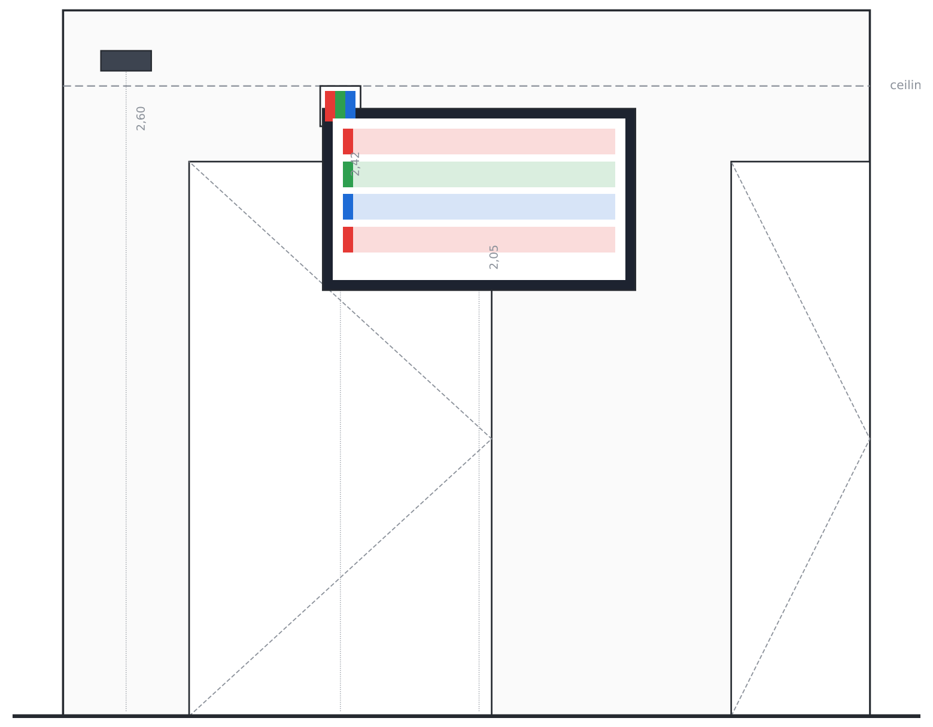
A · TOILET BLOCK – EQUIPMENT & WIRING · 1:30



F · NURSE STATION + TV · 1:30



G · TV ELEVATION (wall P3–P4)



View from the south

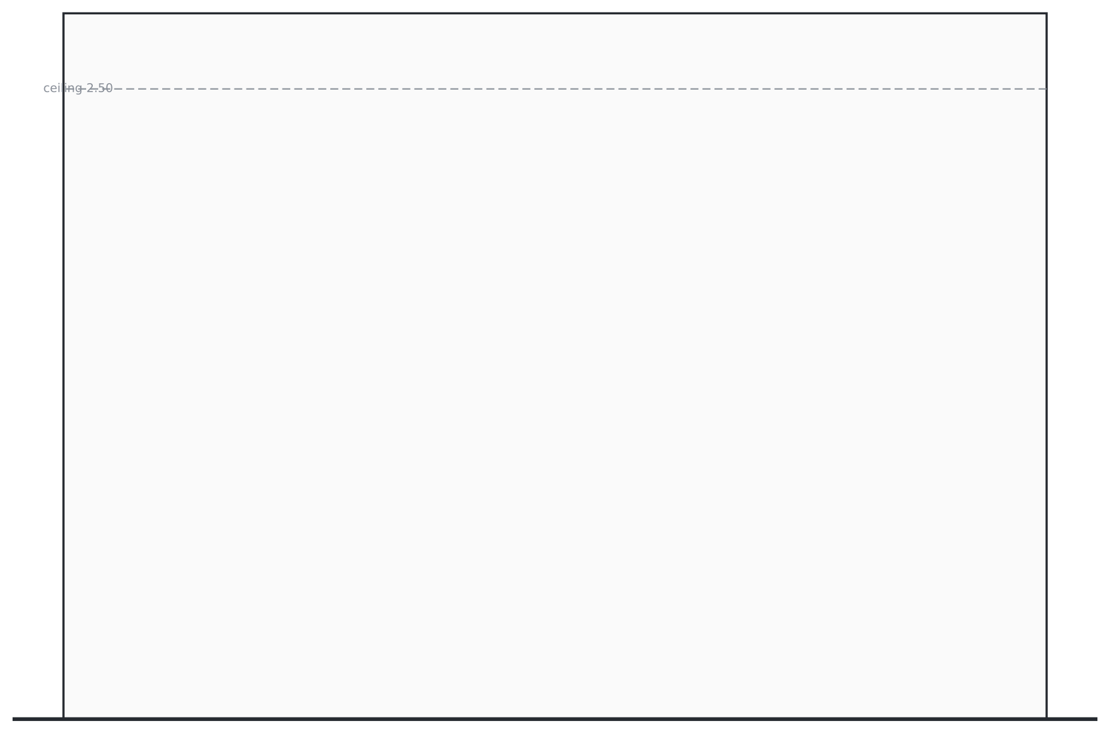
NOTES

1. B1 staff WC (no call).
2. B2: toilet with fixed and fold-down grab bar; pull-cord TI at 1.85 m.
3. B3 accessible: toilet + shower with fold-down seat and grab bars; pull-cords TI (toilet) and TD (shower) at 1.85 m.
4. In the corridor, next to the entrance: LP-B2 / LP-B3 at 2.42 m and PR-B2 / PR-B3 at 1.35 m.
5. RC-B2 and RC-B3 in the corridor ceiling; they join the bus chain (1 and 2).
6. Station with its own door; desk against the corridor wall (window): the nurse faces the corridor and the TV. Phone ACE52190200 / PC ACE52040208.

B · ELEVATION B2 – toilet pull-cord

View from the east

C · ELEVATION B3 – toilet pull-cord

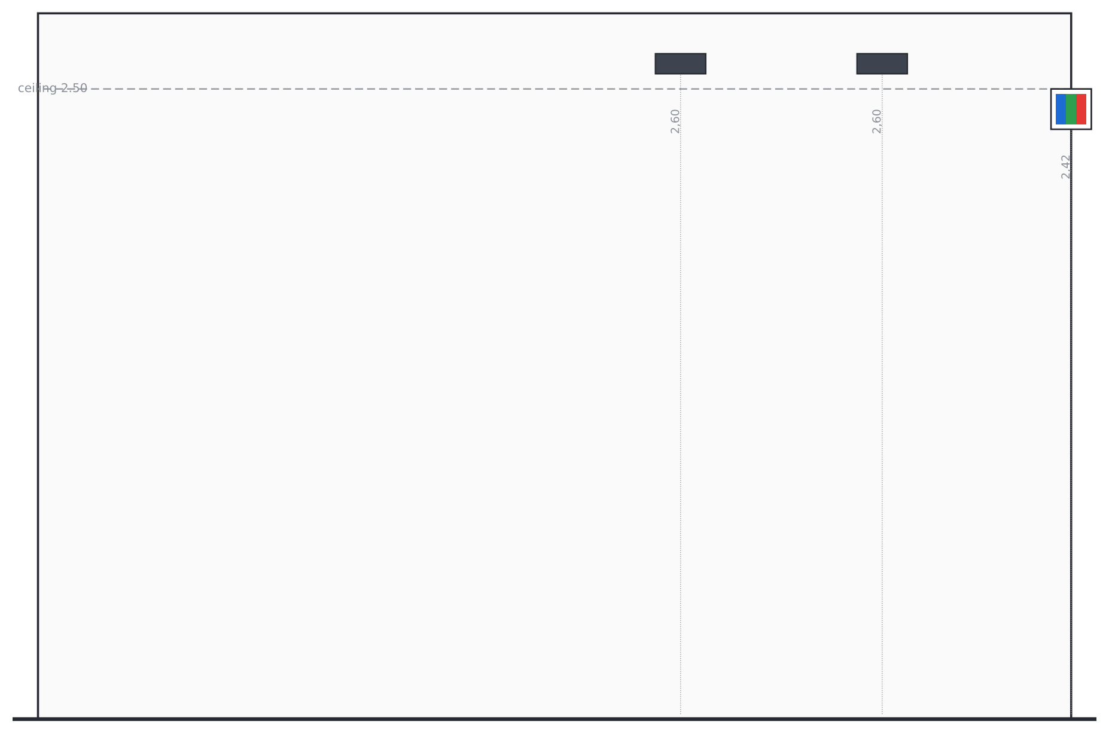


View from the west

D · ELEVATION B3 – shower pull-cord

View from the east

E · CORRIDOR ELEVATION – toilet entrance



View from the north

Pull-cords TI/TD mounted at 1.85 m (≥ 1.80 m) with cord down to 0.20 m from the floor · LP at 2.42 m · PR at 1.35 m · heights in m from finished floor.

INPATIENT WARD · NURSE CALL SYSTEM

Sistema de llamada de enfermería · Pabellón de internación

System: ACE BG · Advantecnia   Prepared by: Telab International

TOILETS & NURSE STATION

Baños y estación de enfermería

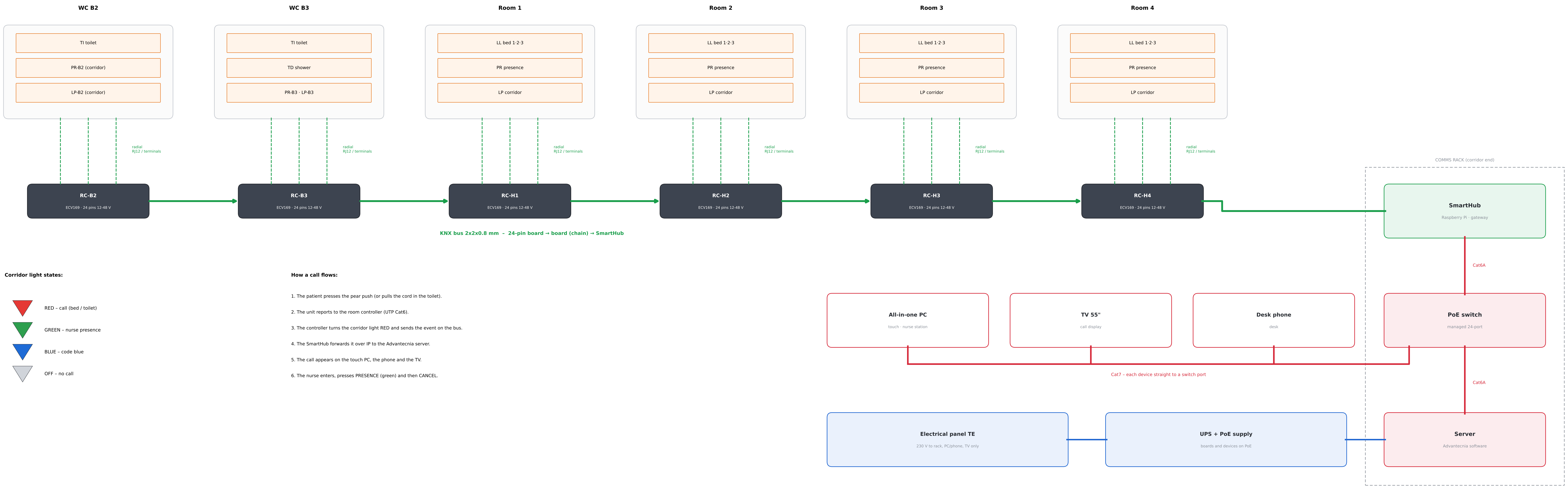
שירותים ועמדת אחיות

SCALE	DATE	REV	SHEET
1:30	12/10/2026	A	04

Dimensions in metres

A1 format · Rev. A – Preliminary design

A · NURSE CALL SYSTEM TOPOLOGY



Corridor light states:

- ▼ RED – call (bed / toilet)
- ▼ GREEN – nurse presence
- ▼ BLUE – code blue
- ▽ OFF – no call

How a call flows:

1. The patient presses the pear push (or pulls the cord in the toilet).
2. The unit reports to the room controller (UTP Cat6).
3. The controller turns the corridor light RED and sends the event on the bus.
4. The SmartHub forwards it over IP to the Advantecnia server.
5. The call appears on the touch PC, the phone and the TV.
6. The nurse enters, presses PRESENCE (green) and then CANCEL.

B · BILL OF MATERIALS

Code	Description	Qty	Remarks
ADVANTECNIA / ACE BG			
LL	Bed call unit with pear-push socket – ACE41220316	12	1 per bed
—	Pear push (cable + button)	12	1 per bed
PR	Presence / cancel – ACE80081001	6	4 rooms + 2 WCs
LP	3-colour corridor indicator light – ACE41020309	6	4 rooms + 2 WCs
TI/TD	Toilet / shower pull-cord call – ACE40320303	3	B2 ×1 · B3 ×2
RC	Converter board / room controller PCB 24 pins 12-48 V ECV169	6	H1 H2 H3 H4 B2 B3
SH	SmartHub (Raspberry Pi) – bus/IP gateway	1	in rack
SV	Server + Advantecnia software – ACE51010103	1	in rack
NETWORK & IT			
PC	Touch PC / monitor – ACE52040208	1	nurse station
TEL	Nurse desk phone – ACE52190200	1	alternative to the PC
TV	TV 55" + wall bracket	1	wall P3–P4
SW	Switch – ACEUCM6302	1	in rack
GAB	42U rack + patch panel + PDU + UPS + PoE supply	1	corridor end
CABLES (calculated length + 15 % spare)			
—	KNX bus 2x2x0.8 mm LH EIBBUS (green)	45 m	calc. 35.8 m
—	Radial call cable device → board (green)	190 m	calc. 161.0 m
—	Cat6 / Cat7 – SmartHub, switch, PC/phone, TV (red)	65 m	calc. 52.6 m
—	Power 3x2.5 mm^2 LSZH (blue)	45 m	calc. 35.3 m
HOSPITAL INFRASTRUCTURE (other trades)			
CT	Bed-head unit (O_2/air/vacuum, 4×230 V, 2×RJ45, light)	12	1 per bed
—	Hospital bed + bedside cabinet + overbed table	12	
—	U curtain track + flame-retardant curtains	4 sets	1 per room
—	Fixed / fold-down grab bars, shower seat	1 lot	B2 · B3

NOTES

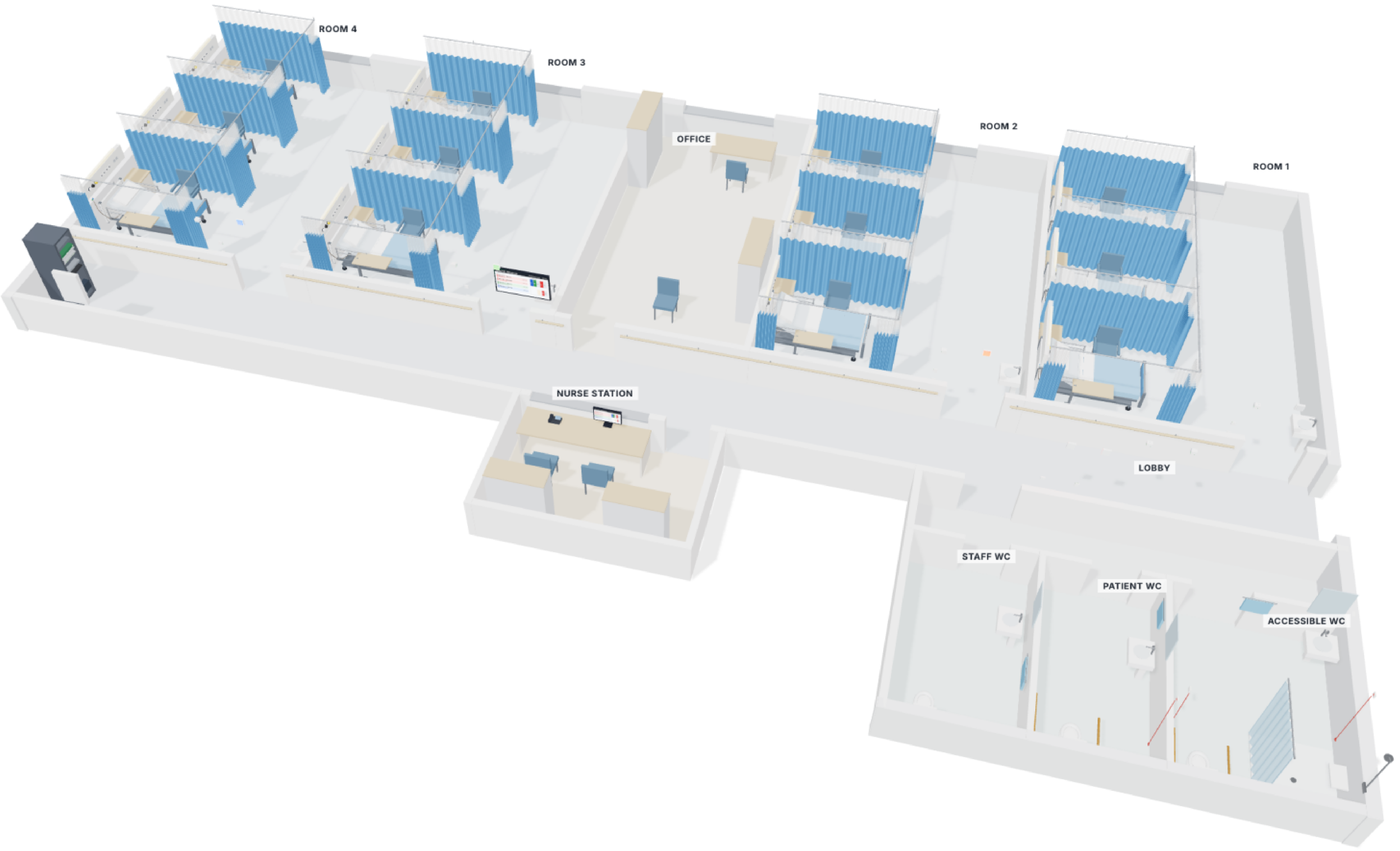
1. Lengths measured on the model routes (vertical drops included) + 15 %.
2. 24-pin boards and call devices are powered by PoE from the rack: no power circuits in the rooms.
3. Confirm with Advantecnia the maximum number of controllers per bus line and total length.
4. Preliminary quantities: verify on site.

INPATIENT WARD · NURSE CALL SYSTEM
Sistema de llamada de enfermería · Pabellón de internación
System: ACE BG · Advantecnia   Prepared by: Telab International

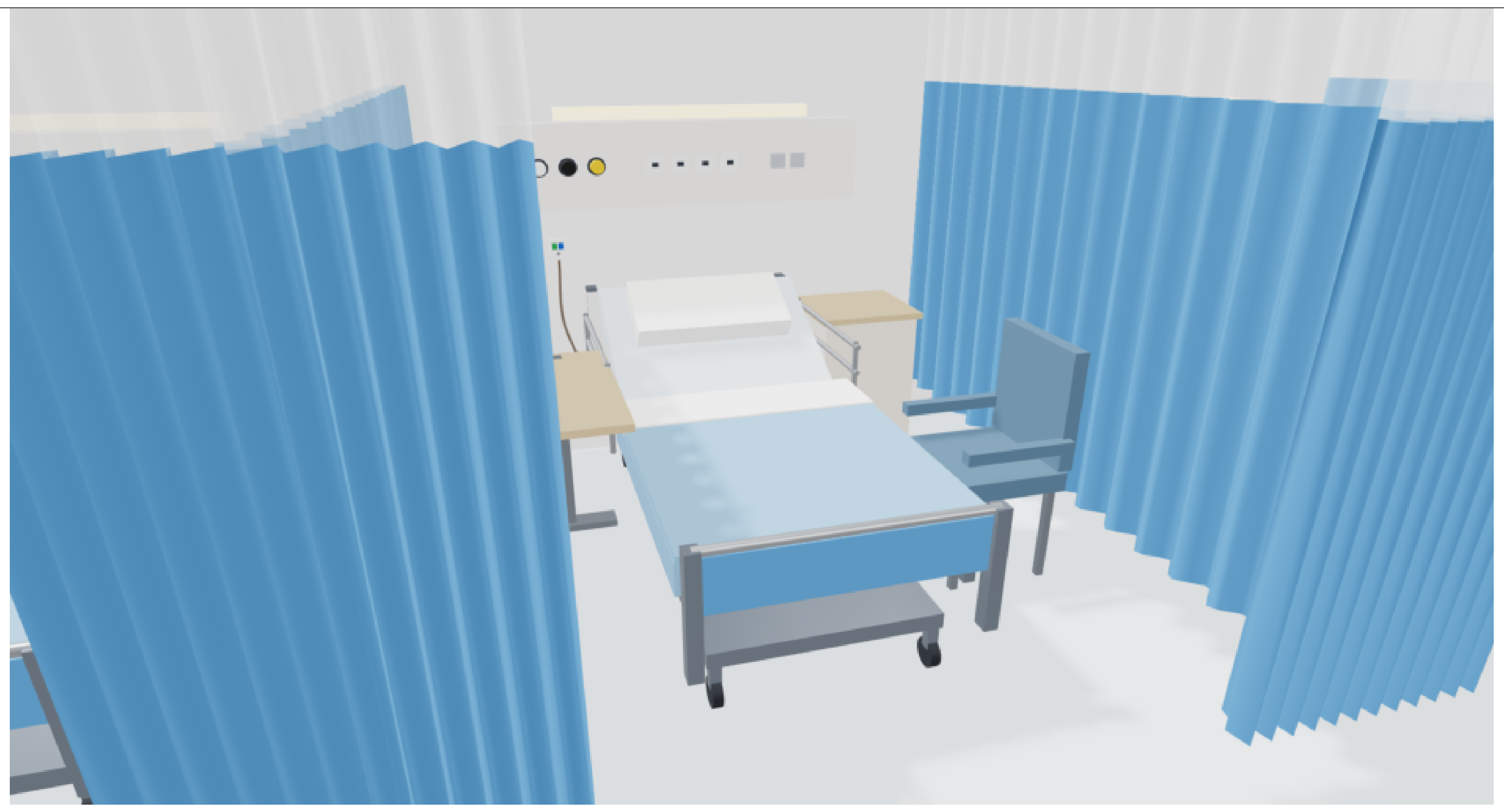
SYSTEM TOPOLOGY & BILL OF MATERIALS
Topología y lista de materiales
טופולוגיית מערכת ורשימת חומרים

SCALE	DATE	REV	SHEET
S/E	**12/10/2026**	**A**	**05**

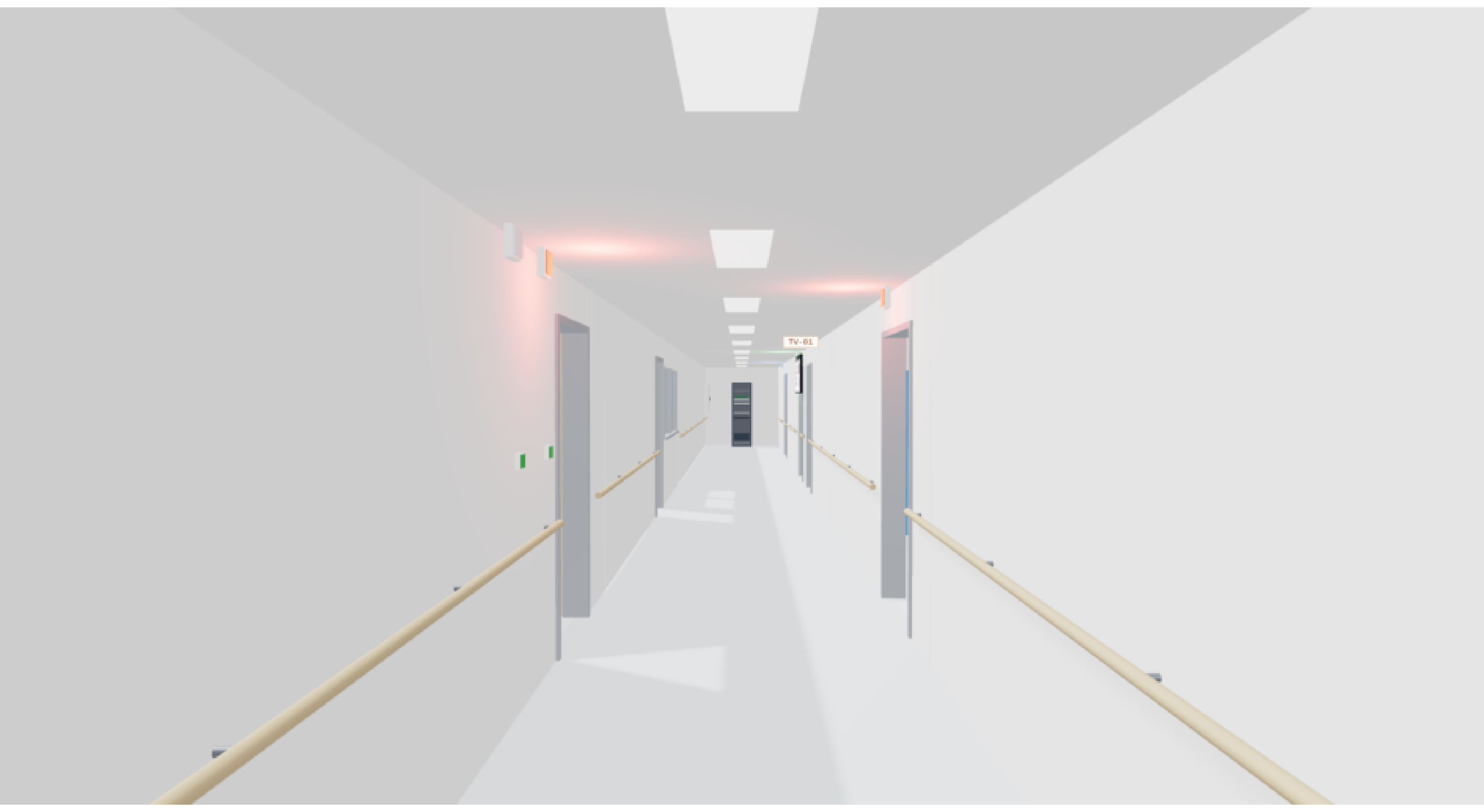
Dimensions in metres
A1 format · Rev. A – Preliminary design



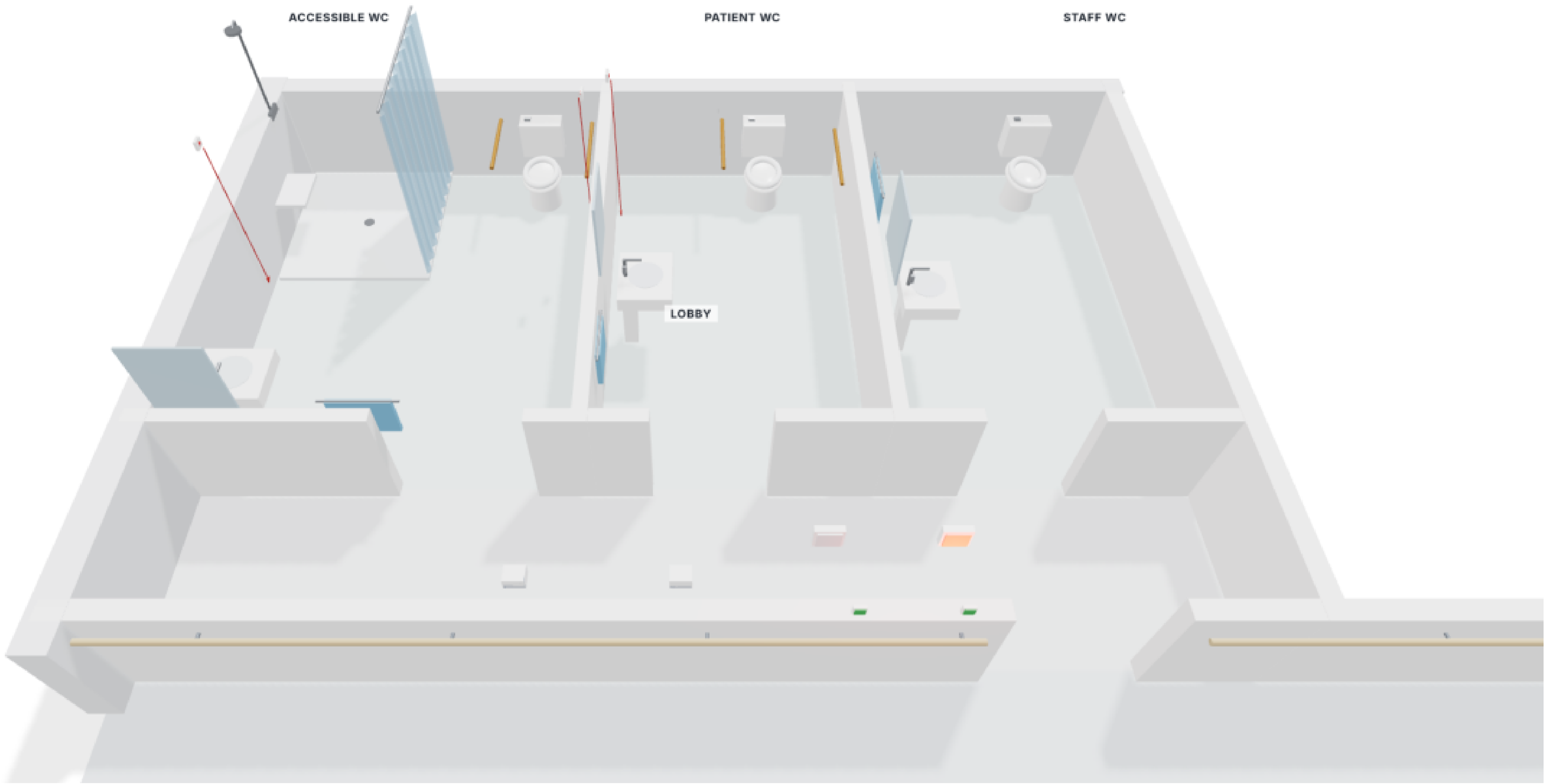
A · Overall isometric - equipment



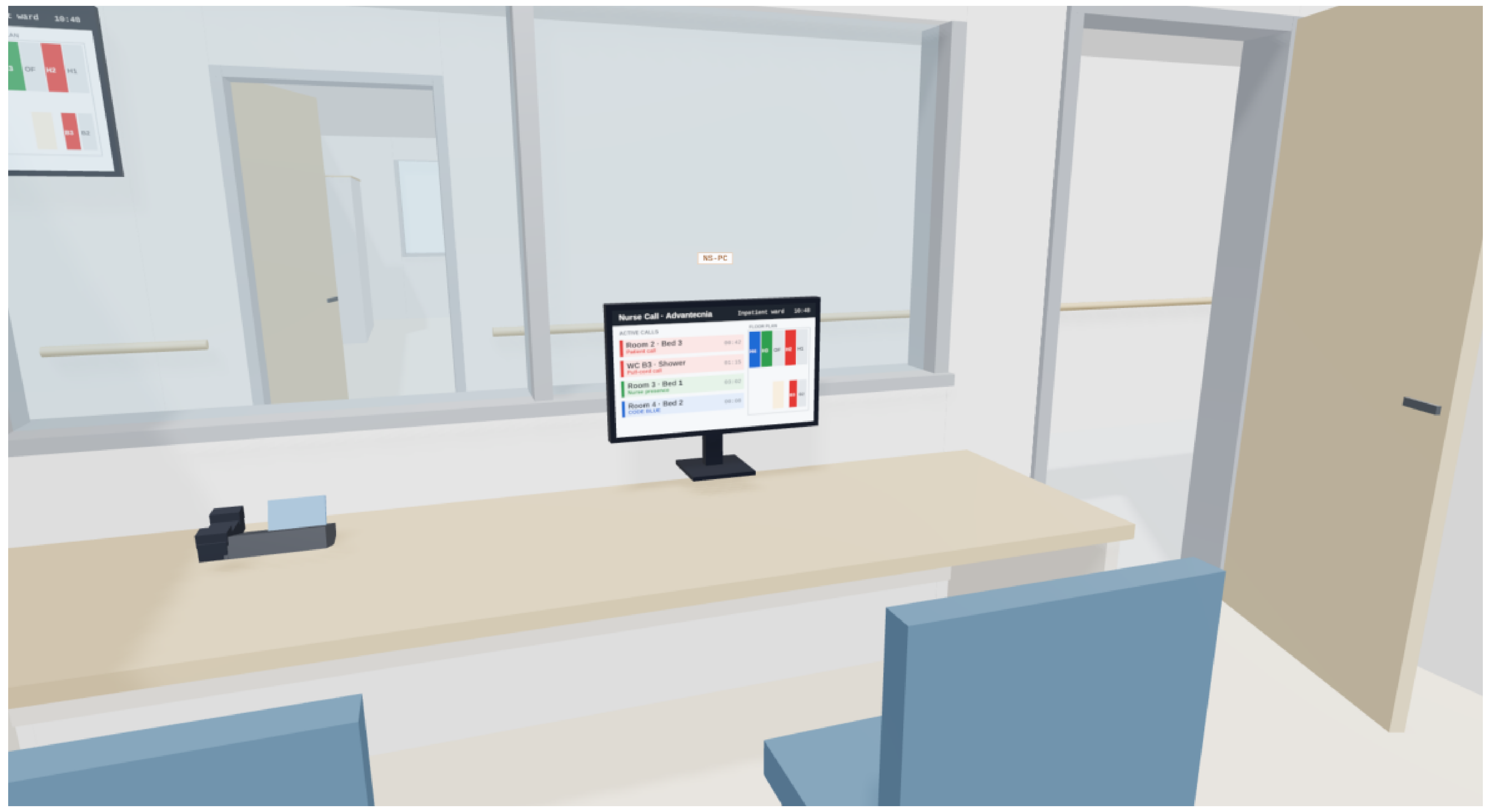
B · Room: beds, curtains, bed-heads and panels



C · Corridor: toilet lights and presence buttons



D · Toilet block



E · Nurse station – Advantecnia touch PC

F · Comms rack

INPATIENT WARD · NURSE CALL SYSTEM

Sistema de llamada de enfermería · Pabellón de internación

System: ACE BG · Advantecnia   Prepared by: Telab International

3D VIEWS – EQUIPMENT

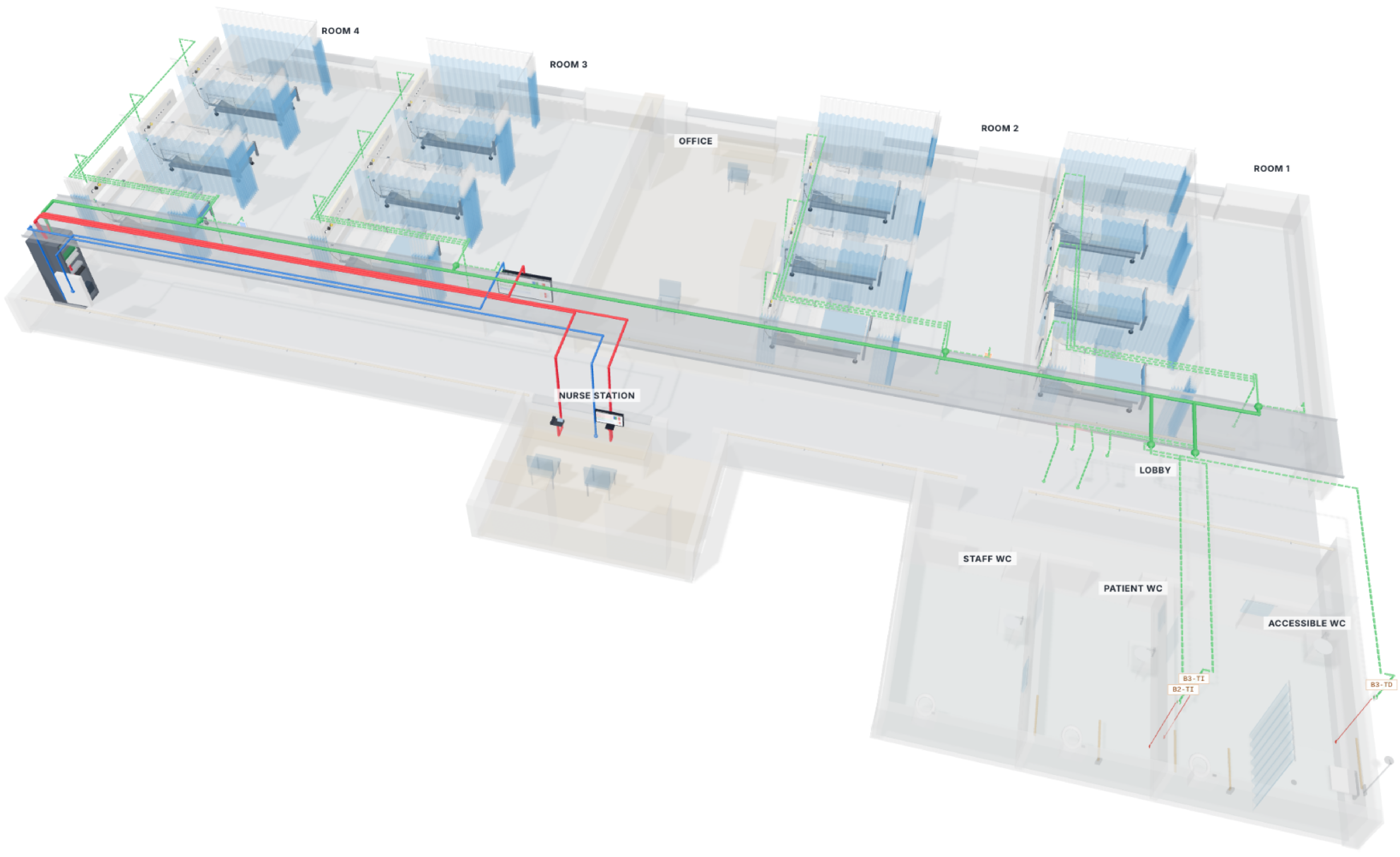
Vistas 3D – equipamiento

ציוד – ממדיים-תלת מבטים

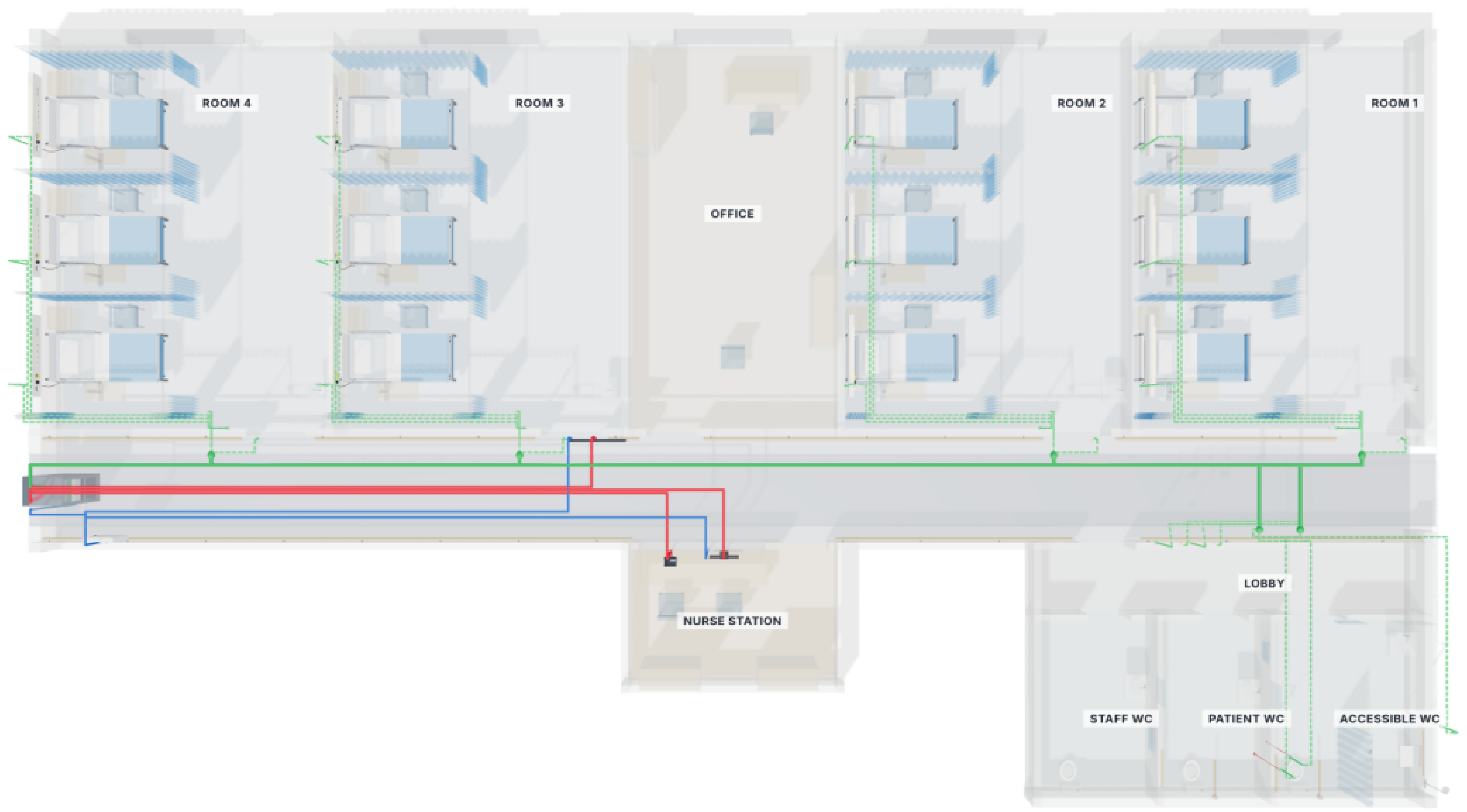
SCALE	DATE	REV	SHEET
S/E	12/10/2026	A	06

Dimensions in metres

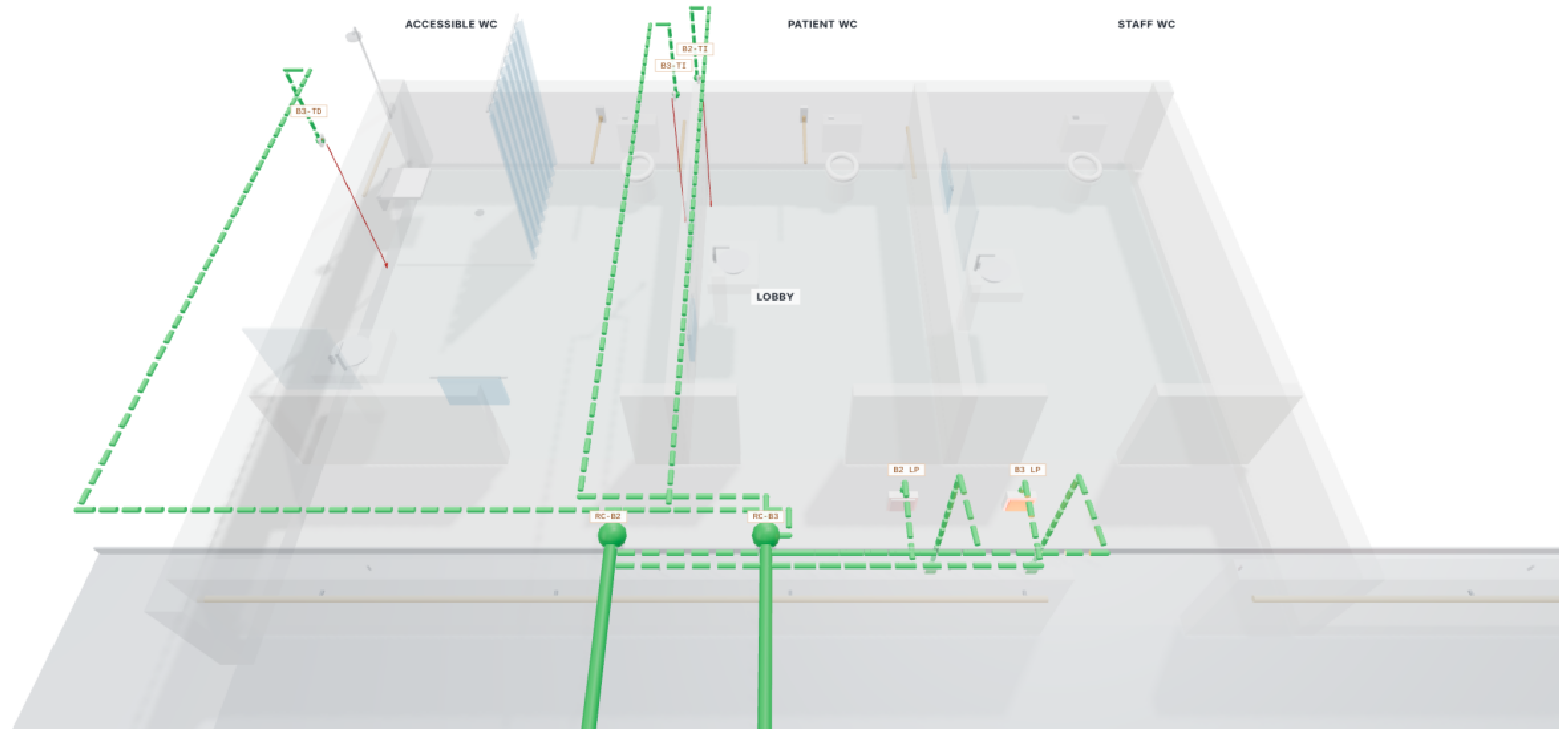
A1 format · Rev. A – Preliminary design



A · Overall isometric – wiring (green: KNX bus / UTP · red: IP network · blue: 230 V)



B · Plan – wiring



C · Toilets – pull-cords, lights and presence to their RC

COLOURS

🟩	KNX bus 2x2x0.8 mm – 24-pin board → board (chain) → SmartHub KNX bus – board-to-board chain · BUS – שרשור כרטיסים
▬ ▬ ▬	Radial in each room – every device to its 24-pin board Radial in each room – device to its 24-pin board · רדיאלי בכל חדר – התקן לכרטיס
🟥	Cat6 / Cat7 only: SmartHub → switch → server · PC / phone · TV Cat6/Cat7 only: SmartHub, switch, server, PC/phone, TV · Cat6/Cat7 בלבד: SmartHub, מתג, שרת, מחשב, TV
🟦	230 V power – rack, PC / phone and TV only (everything else is PoE) 230 V only to rack, PC/phone, TV – rest is PoE · 230V לארון, מחשב/טלפון ו-TV בלבד – השאר PoE

INPATIENT WARD · NURSE CALL SYSTEM
Sistema de llamada de enfermería · Pabellón de internación
System: ACE BG · Advantecnia Prepared by: Telab International

3D VIEWS – WIRING
Vistas 3D – cableado
מבטים תלת-ממדיים – חיווט

SCALE	DATE	REV	SHEET
S/E	**12/10/2026**	**A**	**07**
Dimensions in metres A1 format · Rev. A – Preliminary design			